

Day 173/365

What is a Random Forest?

Many trees voting beats one tree
guessing.

First, the problem with one tree

In my last three posts we built decision trees. They're easy to read, but they have a weakness:

They memorize. A deep tree can fit the training data perfectly, then stumble on new data. Student B from my Day 168 post.

They're jumpy. Change a few rows of data and you can get a quite different tree.

So what if, instead of trusting one tree, we trusted **many**?

The idea: ask a crowd

Imagine a jar of candies. You ask 100 friends to guess how many are inside.

1 friend

Might guess 120. Or 400.
Could be far off.

100 friends

The wild guesses cancel out. The average usually lands close.

A random forest does exactly this with trees. Each tree is a "friend" who gives an opinion. The forest combines them.

Combining many models is called ensemble learning.

Ingredient 1: each tree sees different rows

If every tree saw the same data, they'd all be the same tree. So each one gets its own random sample.

Example: you have 1,000 customers. Tree 1 draws 1,000 rows **with replacement**, like pulling names from a hat and putting each one back.

Some customers show up twice or more.

Some customers are left out. Roughly one-third of them.

This trick is called **bagging**. The left-out rows even act as a free test set later.

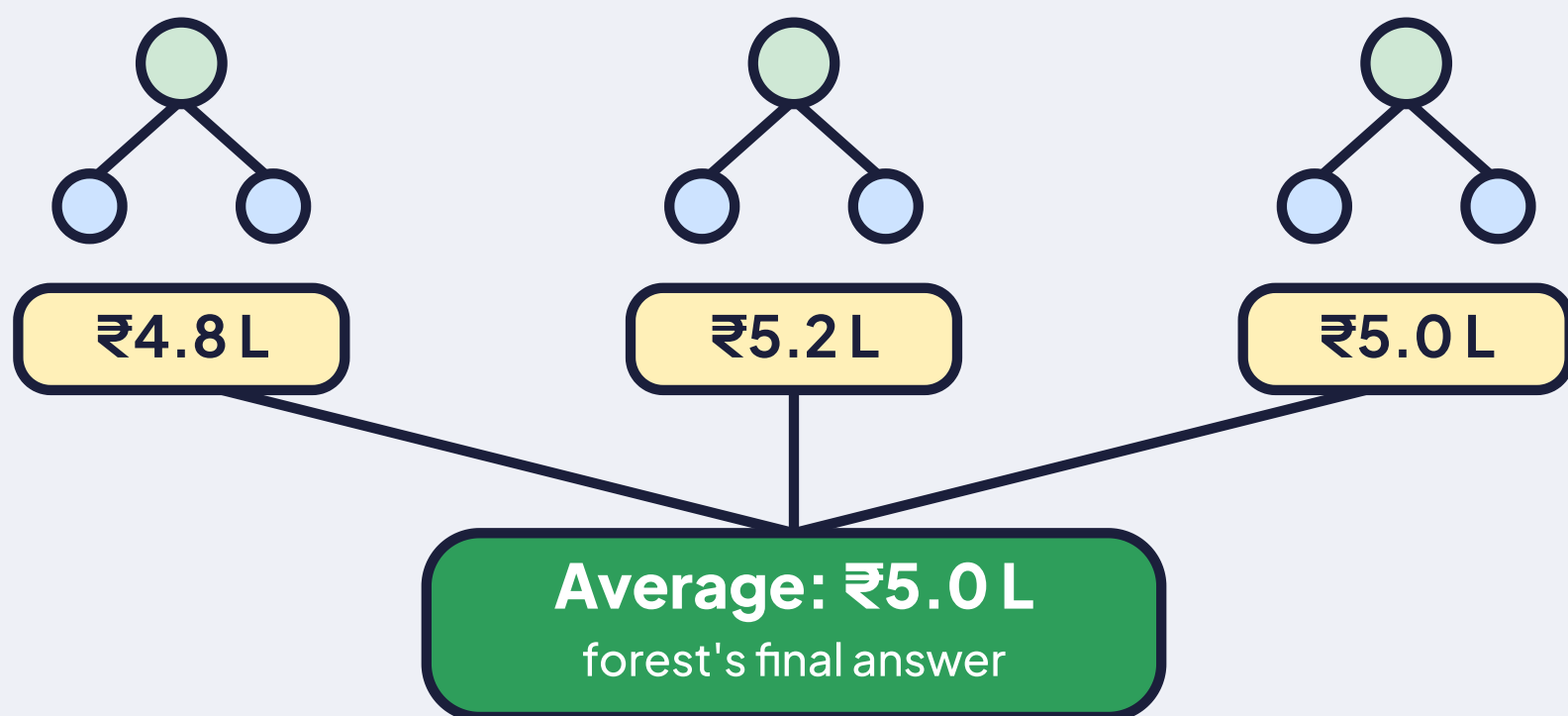
Ingredient 2: each split sees only some features

The problem: if "Bought before?" is by far the strongest feature, every tree will pick it first. The trees end up looking alike again.

The fix: at every split, a tree is only allowed to look at a random handful of features. Sometimes "Bought before?" isn't on the list, so the tree has to find another useful question.

More variety means the trees make **different mistakes**, and different mistakes cancel out when you combine them.

How the forest decides



Predicting a number
Take the **average**, as above.

Predicting a category
Take a **vote**. 70 trees say Buy, 30 say Don't, so Buy.

Illustrative prices for a used car, 3 of 100 trees shown.

Single tree vs. random forest

	One tree	Random forest
Easy to read	Yes	Harder
Memorizes training data	Often	Much less
Stable when data changes	Jumpy	Steadier
Needs feature scaling	No	No
Speed and memory	Light	Heavier

The trade: you give up a bit of readability and speed, and in return you get steadier predictions.

It handles both regression and classification.

Takeaways

A random forest is a crowd of different trees, averaged or voted together.

Different rows for each tree, different features at each split.

Average for numbers, **vote** for categories.

Next: Random Forest Regression in Python, where we'll see it work on real data.